

A Proof that $3 +_s (2 +_s 1) \Downarrow 6$

$$\begin{array}{c}
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 (B-ADD) \frac{}{3 \Downarrow 3} \quad (B-ADD) \frac{(B-ADD) \frac{(B-ADD) \frac{}{2 \Downarrow 2} \quad (B-ADD) \frac{}{1 \Downarrow 1}}{2 +_s 1 \Downarrow 3}}{2 +_s 1 \Downarrow 3} \\
 \\
 (B-ADD) \frac{}{3 +_s (2 +_s 1) \Downarrow 6}
 \end{array}
 \end{array}$$

This is the *derivation tree* of $3 +_s (2 +_s 1) \Downarrow 6$.

As you see, the derivation tree requires the rule (B-NUM) at the leaves.